

Why is it problematic?

It violates the fundamental human right to privacy and cognitive liberty. Treating people as data commodities to be tracked and manipulated, strips away individual autonomy and exploits vulnerabilities for financial gain.

(3) Intellectual Property Exploitation for AI Training

Scenario:

A tech company scrapes millions of copyrighted books, digital artworks, and journalistic articles from the internet without paying or asking permission from the creators. They use this material to train a Gen. AI tool that can now replicate those creators' styles, effectively competing with the original artists.

Why is it problematic?

It bypasses fair compensation and intellectual property rights. By using creative labour without consent to build commercial tools that threaten the livelihoods of those very creators, it violates basic principles of fairness and economic justice.

31. ~~20~~. Data Governance is a collection of processes, roles, policies, standards, and metrics that ensures an organization's data assets are managed securely, accurately, and efficiently. It establishes who has the authority and control over data assets, and how those data assets can be used.

*Key components:

- Data quality management: sets the standards to ensure data is accurate, complete, consistent, and up to date across the entire organization.
- Data security and privacy: Establishes access controls, encryption standards, and compliance rules to protect sensitive info and meet legal mandates like GDPR or POPIA.
- Data architecture and Ownership: Clearly defines who owns specific datasets and assigns accountability for managing day-to-day data tasks.
- Metadata management: Creates a data catalog that defines what different data items mean, where the data is stored, and where it came from.

32. Bias in AI refers to a situation where an artificial intelligence model produces systematically skewed, unfair, or prejudiced results.

An AI model is biased because it mirrors, amplifies, or codifies flaws embedded within its development lifecycle. AI systems do not possess independent moral judgement; they are mathematical engines that find patterns in the data they are fed. If the training data reflects historical human prejudices, lacks diverse representation, or if the developer introduces flawed assumptions during the design phase, the model will treat those systemic inequalities as absolute mathematical rules.

* A real-world example: Flawed Facial Recognition

A well-documented real-world example is the bias found in commercial facial recognition software. Landmark computer science research (such as Gender Shades study) revealed that several prominent facial recognition systems exhibited severe demographic differences. While the technology achieved near-perfect accuracy when identifying lighter-skinned males, its error rates skyrocketed to nearly 35% when attempting to identify darker-skinned females.

This bias occurred primarily because the training datasets used to build the software consisted overwhelmingly of images of white individuals, leaving the AI poorly equipped to recognize diverse facial features.

When biased facial recognition or matching algorithms are deployed in high-stakes environments, the real-world harms or consequences are severe. They could be wrongful arrests and legal injustice, flawed security and border screening, and erosion of public trust.

COMPARISON TABLE

Feature	Data Analyst	Data Scientist	Data Engineer	AI Engineers
Primary Role	Interprets data to solve business problems and create reports.	Builds predictive models and uncovers deep insights.	Constructs and maintains data pipelines and architectures.	Integrates AI models and LLMs into production software.
Core focus	Business intelligence and business analysis.	Advanced statistics, prototyping, and machine learning.	Data infrastructure, scaling, and data cleaning.	Model deployment, software engineering, and system scale.
Key skills	Data storytelling, dashboarding, basic SQL, and business acumen.	Advanced math, stats, machine learning, and experimentation.	Database design, system architecture, and performance optimization.	Software architecture, API integration, and model fine-tuning.
Common Tools	SQL, Excel, Power BI, Tableau.	Python, R, Jupyter, Scikit-learn, Pandas.	SQL, Apache Spark, Airflow, Docker, AWS, Azure/CCP.	Python, PyTorch, Hugging face, LangChain, APIs.

34. Business Intelligence (BI) is a technology-driven process that analyzes raw data and transforms it into actionable, easy-to-understand insights to help business leaders make informed strategic and operational decisions.

* Microsoft Power BI

- A cloud-based business analytics service that integrates deeply with Microsoft systems to connect, clean, and visualize diverse corporate data.

* Tableau

- A powerful data visualization platform known for its ability to handle massive, complex datasets and generate highly interactive, customizable dashboards.

They help businesses make data-driven decisions by consolidating disparate data sources, replacing static reports with interactive dashboards, surfacing trends and anomalies in real time and democratizing access.

35. Concepts in both sections A and B are quite interesting to me. It made realize how much we actually underestimate Data analytics yet we use/work with data on a daily basis and how much deeper one can go to use AI instead of using it in a manner that is "Basic".

36. Python concept. Although I find it interesting, there are certain things I did not know and understand about it. So I had to do more Google work and Youtube.

37. I am an ~~Ex~~ currently working as an Exploration Geologist and I work with data everyday. This will help me optimize some of my functions at work. I am also a forex trader and this will help me derive all those insights I could be missing in the data that I journal.

References

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